

SMAR 2026 - 8th International Conference on Smart Monitoring, Assessment and Rehabilitation
of Civil StructuresFleet monitoring of bridges using drive-by operational deflection
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Abstract

Operational deflection shapes (ODS) provide valuable spatial insight into bridge dynamics and can support damage assessment. This study proposes a drive-by framework for estimating ODS-related information using vehicle-mounted accelerations only. A transfer function based approach is used to reconstruct contact-point bridge acceleration, followed by band-pass filtering and Hilbert transform based amplitude extraction to obtain spatial response envelopes along the bridge. The method is evaluated using a coupled vehicle bridge interaction model under both smooth and stochastic Class A road profiles, considering healthy and locally damaged cases. A Monte Carlo study with a fleet of vehicles is conducted to assess robustness to variability in vehicle properties. Results show that the reconstructed responses capture the dominant modal behaviour and produce consistent amplitude patterns across the fleet. While differences between healthy and damaged cases are subtle after normalisation, combining front and rear axle responses improves stability and repeatability. The findings demonstrate the potential of drive-by monitoring for extracting ODS-related information towards scalable bridge assessment.

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1. Introduction

Bridge infrastructure plays a critical role in modern transport networks, and ensuring its safety requires reliable and efficient condition assessment (Khan et al., 2016). Conventional monitoring methods typically rely on fixed sensors installed on the structure, such as accelerometers and strain gauges (Moreno-Gomez et al., 2018; Huang et al., 2022; Zhang et al., 2022). While these systems provide high-quality data, their widespread deployment is limited by installation cost, maintenance requirements, and scalability challenges. As a result, increasing attention has been given to indirect or drive-by monitoring approaches, where instrumented vehicles are used to extract bridge-related dynamic information during normal crossings (Li et al., 2026). Drive-by monitoring offers a low-cost and scalable alternative; however, the measured vehicle response is influenced not only by the bridge but also by vehicle dynamics, tyre-road interaction, axle configuration, and road roughness (Hester & González, 2017; Li et al., 2023; Lan et al., 2023). These factors can obscure structural contributions and complicate the identification of reliable damage-sensitive indicators, particularly for localised damage (Carnevale et al., 2019). Operational deflection shapes (ODS) provide valuable spatial information on structural behaviour under operational loading and have been widely used in conventional structural health monitoring (OBrien et al., 2023). However, existing ODS estimation methods typically require direct bridge instrumentation, limiting their applicability at network scale. Developing a drive-by framework capable of estimating ODS-related information from vehicle-only measurements is therefore of significant interest.

This study proposes an acceleration-based drive-by method to estimate bridge-related ODS information using a two-axle vehicle. A transfer function based formulation is used to reconstruct contact-point bridge acceleration directly from vehicle acceleration measurements. The reconstructed response is first processed using band-pass filtering to mitigate the influence of road profile effects and subsequently analysed using a Hilbert transform (HT) based approach to extract the amplitude envelope associated with the dominant bridge mode. The method is evaluated using a coupled vehicle bridge interaction model, considering both smooth and stochastic Class A road profiles and localised damage scenarios. In addition to single-vehicle simulations, a Monte Carlo fleet analysis is performed with stochastic variations in vehicle properties to assess robustness. The study aims to evaluate (i) the accuracy of contact-point reconstruction, (ii) the effectiveness of HT-based amplitude extraction for distinguishing healthy and damaged cases, and (iii) the robustness of the approach under vehicle variability, contributing towards scalable drive-by bridge monitoring.

2. Vehicle Bridge Interaction Modelling Framework

The vehicle is modelled as a four-degree-of-freedom half-car system consisting of a sprung mass representing the vehicle body and two unsprung masses representing the front and rear axles. The bridge is represented as a simply supported Euler-Bernoulli beam with a span length of 20 m, discretised into 20 finite elements of 1 m length, with each node possessing vertical displacement and rotational degrees of freedom. Simply supported boundary conditions are applied by restraining the vertical displacement at both end nodes while leaving the rotational degrees of freedom free. The bridge width and depth are 15 m and 1 m, respectively, corresponding to a span-to-depth ratio of 20. A material density of 2500 kg/m³ and Young's modulus of 35 GPa are adopted, giving a flexural rigidity of 4.375×10^{10} N·m², while structural damping is neglected. The mean vehicle parameters include a sprung mass of 16200 kg, front and rear unsprung masses of 700 kg and 1100 kg, a pitch inertia of 93457 kg·m², an axle spacing of 4.75 m, suspension stiffnesses of 400 kN/m and 1000 kN/m, suspension damping coefficients of 10 kN·s/m and 20 kN·s/m, tyre stiffnesses of 1750 kN/m and 3500 kN/m, and tyre damping coefficients of 72 kN·s/m for both axles. To simulate realistic fleet variability, all vehicle parameters are sampled from truncated normal distributions with standard deviations equal to 20% of the corresponding mean values and bounded within ± 2 standard deviations. The interaction between the vehicle and bridge is established through tyre contact forces governed by the relative displacement and velocity between the vehicle axles, bridge surface, and road profile. The road profile is modelled as a stochastic process based on a prescribed power spectral density, with both a standard ISO Class A profile (ISO, 2016) and an idealised smooth profile considered. Local structural damage is introduced through a spatial reduction in bridge flexural rigidity following (Sinha et al., 2002). A detailed description of the VBI modelling framework can be found in (González et al., 2010), while the adopted model parameters are based on (Feng et al., 2023).

3. Methodology

3.1. Acceleration-based transfer function

The contact-point response is reconstructed using an acceleration-based transfer function approach, where vehicle accelerations and known vehicle properties are used as inputs. The vertical accelerations of the sprung and un-sprung masses, together with pitch motion, are used to estimate contact-point accelerations at tyre-road interfaces. The reconstruction is performed in the frequency domain. Applying the Fourier transform to vehicle equations of motion:

$$A(\omega) X(\omega) = B(\omega) W_c(\omega) \quad (1)$$

where $X(\omega)$ contains the measured vehicle acceleration spectra, $W_c(\omega)$ represents the unknown contact-point accelerations, and $A(\omega)$ is the frequency-dependent dynamic stiffness matrix of the vehicle. The matrix $B(\omega)$ contains the tyre dynamic terms relating the contact forces to the contact-point motion. The contact-point accelerations are then obtained directly from the un-sprung-mass equilibrium equations as:

$$\ddot{W}_{c,f}(\omega) = \frac{R_3(\omega)}{B_f(\omega)} \quad (2)$$

$$\ddot{W}_{c,r}(\omega) = \frac{R_4(\omega)}{B_r(\omega)} \quad (3)$$

where $R(\omega) = A(\omega)X(\omega)$, and $B_f(\omega)$ and $B_r(\omega)$ denote the front and rear tyre dynamic impedance terms. The reconstructed contact-point accelerations are transformed back to the time domain using the inverse Fourier transform, with Hermitian symmetry enforced to ensure real-valued signals. A low-pass filter is subsequently applied to remove high-frequency components and improve numerical stability. Overall, this acceleration-based transfer function approach enables direct reconstruction of bridge-related contact-point responses from vehicle measurements, forming the basis for subsequent HT-based amplitude extraction. Further details on the transfer function derivation can be found in previous work (González et al., 2023; Feng et al., 2025).

3.2. Operational deflection shape via Hilbert transform

Following the reconstruction of the contact-point accelerations, the bridge ODS is estimated using a HT-based approach (Kunwar et al., 2013). The bridge-related acceleration responses at the front and rear axles, $\ddot{w}_{br,f}(t)$ and $\ddot{w}_{br,r}(t)$, are first obtained by removing the road-induced component from the reconstructed contact-point signals. To isolate the dominant structural response, the signals are first low-pass filtered to remove high-frequency noise, followed by a band-pass filter centred around the first natural frequency of the bridge, f_1 . The passband is defined as: $f_{\text{low}} = \max(0.01, f_1 - \Delta f)$, and $f_{\text{high}} = f_1 + \Delta f$, where Δf denotes the selected half-bandwidth. The resulting filtered signals are denoted as: $\ddot{w}_{br,f}^{(bp)}(t)$, $\ddot{w}_{br,r}^{(bp)}(t)$. The HT is then applied to obtain the analytic signal:

$$z(t) = x(t) + j \mathcal{H}\{x(t)\} \quad (4)$$

from which the amplitude envelope is defined as: $A(t) = |z(t)|$. The envelopes of the filtered signals are used as approximations of the ODS associated with the dominant mode, giving:

$$ODS_f(t) = |\mathcal{H}\{\ddot{w}_{br,f}^{(bp)}(t)\}| \quad (5)$$

$$ODS_r(t) = |\mathcal{H}\{\ddot{w}_{br,r}^{(bp)}(t)\}| \quad (6)$$

Since the vehicle is in motion, the time-domain responses are mapped to spatial coordinates using $x = vt$, allowing the ODS to be expressed as functions along the bridge span. The front and rear estimates are interpolated onto a common spatial grid, and a combined ODS is obtained as:

$$ODS_{comb}(x) = \frac{1}{2} \left(ODS_f(x) + ODS_r(x) \right) \quad (7)$$

All ODS results are normalised by their maximum value to enable consistent comparison between cases. This HT-based approach provides a simple and effective means of extracting spatial structural information from vehicle-derived signals, while combining front and rear responses improves robustness and reduces axle-specific variability.

4. Results and discussions

4.1. Drive-by monitoring with single vehicle

This section investigates the feasibility of detecting local bridge damage using a single two-axle vehicle. The framework combines transfer function-based reconstruction of the contact-point response with HT-based amplitude extraction under both smooth and standard Class A road profiles. Figures 1 and 2 present the smooth road results. Figure 1 shows strong agreement between the reconstructed and exact contact-point responses, as well as their corresponding normalised HT amplitudes, confirming that the reconstruction preserves key bridge-related dynamic information. Figure 2 compares the normalised amplitudes for healthy and damaged cases. Clear differences are observed, particularly in the rear and combined responses, demonstrating that the extracted HT amplitudes are sensitive to local stiffness changes and can identify damage under ideal conditions.

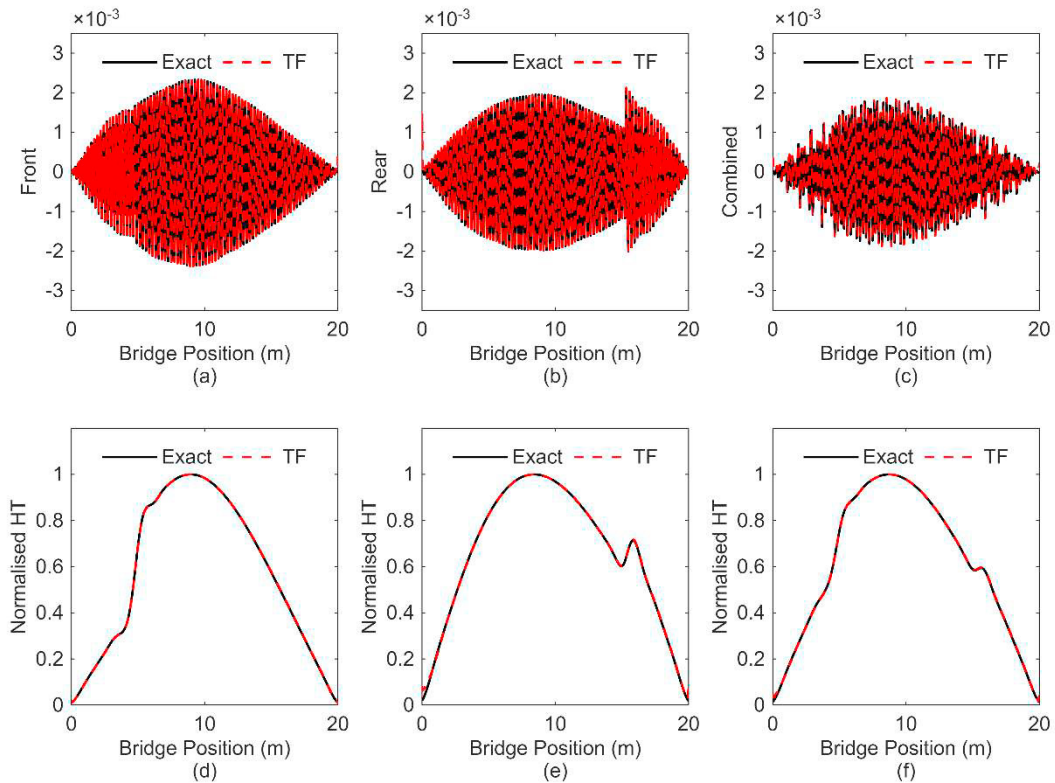


Fig. 1. Comparison between exact and transfer function reconstructed contact-point responses for a smooth road profile. (a-c) front, rear, and combined accelerations show close agreement, while (d-f) normalized amplitudes are consistent overall.

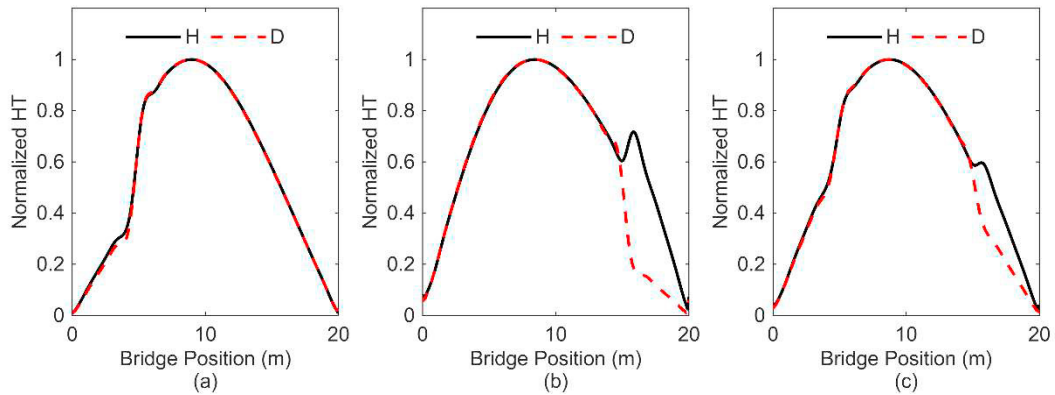


Fig. 2. Comparison of normalized amplitudes for healthy (H) and damaged (D) cases from reconstructed contact-point responses for a smooth road profile. (a-c) front, rear, and combined results.

Figures 3 and 4 extend the analysis to the Class A road profile. Figure 3 shows good agreement between the exact and reconstructed responses and their corresponding amplitudes despite road-induced excitation. Figure 4 compares the healthy and damaged amplitudes under road roughness. Although the responses become more irregular, consistent differences between the two cases remain visible, particularly in the rear and combined results.

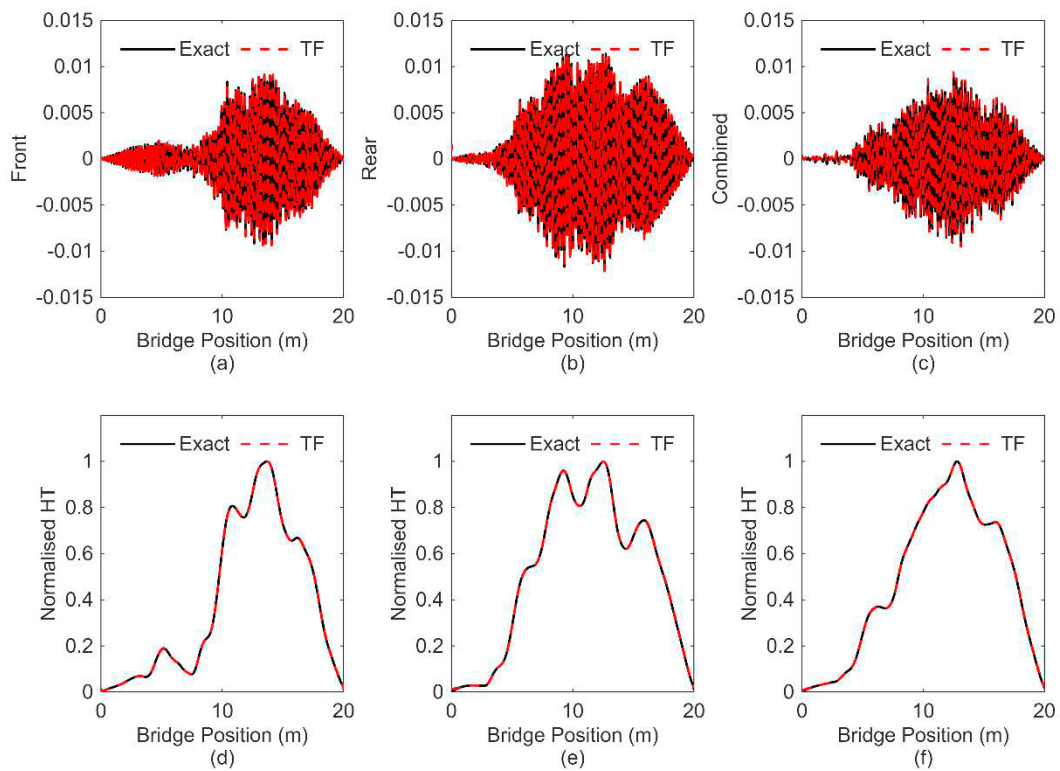


Fig. 3. Comparison between exact and transfer function reconstructed contact-point responses for a standard Class A road profile. (a-c) front, rear, and combined accelerations show close agreement, while (d-f) normalized amplitudes are consistent overall, with a minor deviation in the rear response near the end of the span.

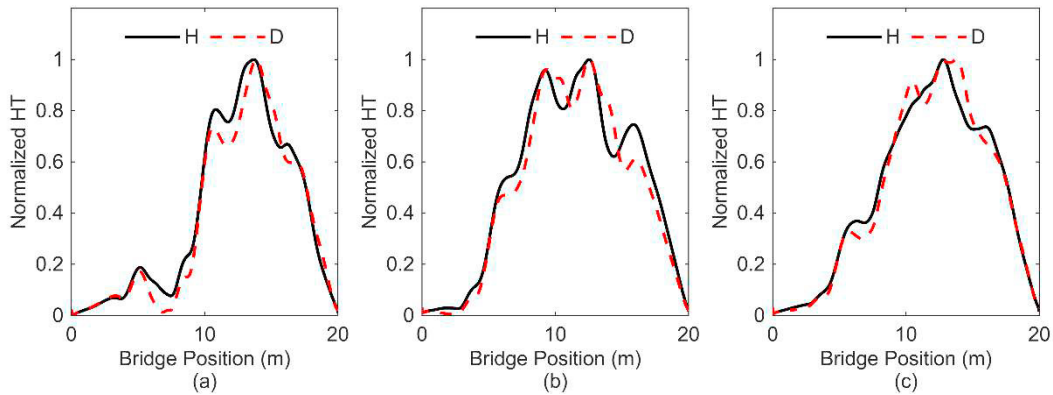


Fig. 4. Comparison of normalized amplitudes for healthy (H) and damaged (D) cases from reconstructed contact-point responses for a standard Class A road profile. (a-c) front, rear, and combined results.

Overall, the results demonstrate that the proposed framework can accurately reconstruct bridge-related responses and identify damage-sensitive differences through normalised HT amplitudes. Although road roughness introduces additional variability and reduces the clarity of the damage signatures, distinguishable differences between healthy and damaged conditions remain observable, particularly in the rear and combined responses. The consistent agreement between exact and reconstructed results under both smooth and rough road conditions further confirms the robustness of the transfer function reconstruction approach. These findings indicate that the proposed framework has strong potential for practical drive-by bridge monitoring applications using vehicle-based measurements alone.

4.2. Drive-by monitoring with a fleet

To evaluate robustness, Monte Carlo simulations were performed using a fleet of 100 two-axle half-car vehicles with key vehicle parameters sampled from truncated normal distributions with 20% coefficient of variation. Coupled vehicle-bridge simulations were conducted for both healthy and 10% local damage cases under a Class A road profile. Contact-point bridge accelerations were reconstructed using the proposed transfer function-based approach. After removing the road-induced component, a band-pass filter and HT were applied to extract normalised spatial amplitude envelopes, which were interpolated onto a common spatial grid. A combined response was obtained by averaging the front and rear axle amplitudes.

Figure 5 presents the mean normalised HT amplitudes together with the corresponding standard deviation bands for the healthy and damaged cases. Although the healthy and damaged responses exhibit similar overall shapes due to the normalisation process, consistent differences in amplitude magnitude and spatial evolution remain observable across the span, indicating that the extracted HT amplitudes retain sensitivity to structural changes despite significant vehicle variability and road-induced excitation. The front-axle response in Figure 5(a) exhibits the largest variability, with considerable overlap between the healthy and damaged standard deviation bands, although local differences in amplitude trends remain visible, particularly near the central region of the bridge span. The rear-axle response in Figure 5(b) provides smoother behaviour and clearer distinction between the two structural states, suggesting improved robustness to random variability. The most stable and consistent performance is achieved in the combined response shown in Figure 5(c), where reduced variability and smoother mean curves allow the healthy and damaged conditions to be more clearly distinguished. Overall, the results demonstrate that normalised HT amplitudes derived from reconstructed contact-point responses can preserve damage-sensitive characteristics under realistic operational conditions. While individual axle responses may contain noticeable scatter due to vehicle variability and road roughness, combining information from multiple axle locations significantly improves stability and enhances the visibility of damage-related features. These findings highlight the strong potential of the proposed framework for practical fleet-based drive-by bridge monitoring applications using heterogeneous vehicle populations under realistic traffic and road conditions.

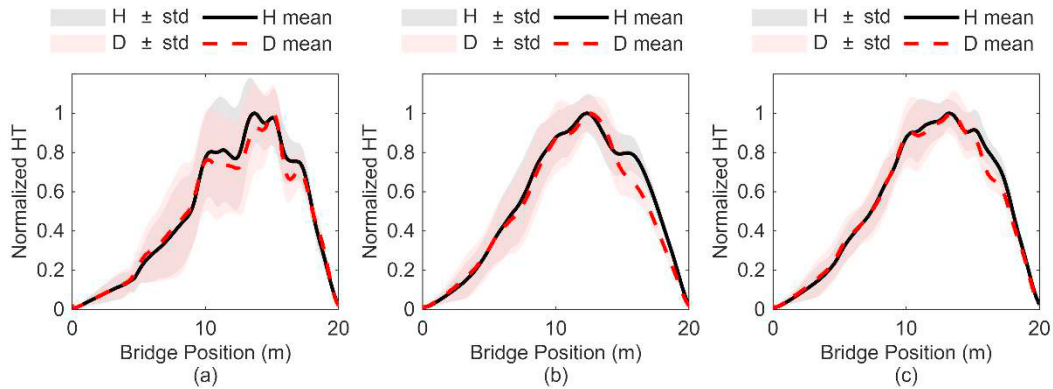


Fig. 5. Comparison of normalized amplitudes for healthy (H) and damaged (D) cases from reconstructed contact-point responses for a standard Class A road profile. (a-c) Front, rear, and combined results, based on Monte Carlo simulations with a fleet of 100 vehicles.

5. Conclusions

This paper presented an acceleration-based drive-by monitoring framework for estimating bridge operational deflection shape information using vehicle-mounted accelerometers only. The method combines a transfer function based approach for reconstructing tyre contact-point accelerations with HT based amplitude extraction to obtain spatial representations of the dominant bridge modal response. The framework was systematically evaluated using a coupled half-car-bridge interaction model under both smooth and stochastic Class A road profiles, considering healthy and locally damaged structural states, and further assessed through Monte Carlo fleet simulations incorporating stochastic vehicle property variability.

The single-vehicle results confirmed that the transfer function reconstruction accurately recovers bridge-related contact-point responses under both road conditions, with close agreement between reconstructed and exact signals. Under smooth road conditions, normalised HT amplitudes clearly differentiated healthy and damaged cases, demonstrating the sensitivity of the proposed indicator to local stiffness reductions. Under Class A road roughness, the responses became more irregular due to road-induced excitation (Feng et al., 2024; Erduran et al., 2025), yet consistent differences between healthy and damaged states remained observable, particularly in the rear and combined axle responses. This confirms that band-pass filtering centred on the bridge fundamental frequency is effective in isolating the dominant structural contribution from road-contaminated signals (Li et al., 2020).

The fleet-based Monte Carlo analysis demonstrated that the proposed framework maintains robustness under realistic vehicle heterogeneity. Although individual axle responses exhibited scatter due to the combined effects of vehicle variability and road roughness, the combined front-rear amplitude response consistently produced more stable and repeatable results across the fleet (Sitton et al., 2020). This highlights the advantage of exploiting multi-axle information in drive-by monitoring, where aggregating spatial responses reduces the influence of vehicle-specific dynamics and improves the reliability of extracted damage-sensitive features. The spatial consistency observed across the simulated fleet further supports the feasibility of deploying such a framework within large-scale bridge networks, where vehicles from normal traffic serve as mobile sensing platforms without requiring dedicated instrumentation on each structure (Li et al., 2026).

Overall, the normalised HT amplitudes derived from reconstructed contact-point responses demonstrate viable sensitivity to structural damage and represent a promising indicator for indirect bridge condition assessment at network scale. The current study is limited to numerical simulations with simplified modelling assumptions and presumes knowledge of the road profile. Future work will address more complex bridge configurations, heterogeneous traffic conditions, and unknown road profiles, alongside the development of more discriminative damage indices and rigorous experimental validation.

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